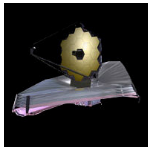

3087 - Reaching ~0.1 arcsec inner working angle for NIRCam coronagraphic imaging

Cycle: 2, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Dr. Bin Ren (PI) (ESA Member)	**Observatoire de la Cote d'Azur**
Dr. Kevin Fogarty (CoI)	NASA Ames Research Center
Dr. John Henry Debes (CoI) (ESA Member)	Space Telescope Science Institute - ESA - JWST
Dr. Chris Stark (CoI)	NASA Goddard Space Flight Center
Dr. Laurent Pueyo (CoI)	Space Telescope Science Institute
Julien Girard (CoI)	Space Telescope Science Institute
Dr. Myriam Benisty (CoI) (ESA Member)	Max Planck Institute for Astronomy
Dr. Chen Xie (CoI) (US Admin CoI) (Contact)	The Johns Hopkins University

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	HR-4796A offset-0	NIRCam Coronagraphic Imaging	(1) HR-4796A
	2	HR-4796A offset-0.1arcsec	NIRCam Coronagraphic Imaging	(1) HR-4796A
	3	HR-4796A offset-0.15arcsec	NIRCam Coronagraphic Imaging	(1) HR-4796A
	4	PSF offset-0	NIRCam Coronagraphic Imaging	(2) HR-4735
	5	PSF offset-0.1arcsec	NIRCam Coronagraphic Imaging	(2) HR-4735
	6	PSF offset-0.15arcsec	NIRCam Coronagraphic Imaging	(2) HR-4735

ABSTRACT

Depending on the physical sizes of the coronagraphic round masks on NIRCam, they can image the surrounding environments of central bright sources down to ~0.4 arcsec inner working angle (IWA). Nevertheless, to image closer-in regions, the roll restriction of JWST poses fundamental limits on the coverage of bar masks. With the established on-sky measurement of the ~1.1 mas pointing stability of JWST, we can push beyond the IWA for the NIRCam round masks.

By performing MASK210R observations with post-TA mask offsets of 0.0", 0.1", and 0.15" (in a single direction), we could image the surroundings of central sources down to 0.1 arcsec along that direction. We propose to perform such an instrumental calibration exploration for the MASK210R occulter of NIRCam, and thus push the supported IWA to one that supersedes the stated values. Using a total of 1.6 hours of on-target exposure for the prototypical debris ring around HR 4796A and its point spread function star, we expect to recover the minor axis of the ring at a radius of ~0.15 arcsec. Upon the completion of this program, we will provide a NIRCam technical report to establish formal guidelines in using this mode for future users. By establishing a ~0.1 arcsec IWA for NIRCam, we will provide the access to close-in planets and close-in substructures (both circumstellar and circum-quasar): all these regions are where we have identified evidence for planet-disk interactions or circumnulear structures, which are hidden with other high-contrast imagers or ALMA but uniquely accessible by JWST.

OBSERVING DESCRIPTION

Using a total of 1.6 hours of on-target exposure for the prototypical, well-characterized, debris ring around HR 4796A and its point spread function (PSF) star, we expect to recover the minor axis of the ring at a radius of ~0.15 arcsec. Upon the completion of this program, we will provide a NIRCam instrumental science report to establish formal guidelines for using this mode by future users.

For HR 4796A, we will observe it with three JWST observations (each with an independent Target Acquisition). In each observation, we apply a single post-TA mask offset of 0.0", 0.1", or 0.15" (all in the same direction, along the disk minor axis). By combining the RDI-subtracted results from the three offsets, we will establish the contrast performance as a function of mask offset and angular separation in this direction. ADI will not be used as a PSF subtraction method for our disk target, since the small field rotation can introduce substantial disk self-subtraction. Instead, our PSF subtraction strategy relies on RDI using a closely matched reference star.

For the reference star HR 4735, it is carefully chosen in existing HST (and confirmed in JWST wavelengths) observations to empirically capture the PSF of HR 4796A. By obtaining an identical post-TA offset strategy (0.0", 0.1", 0.15") and matched instrument setup, we will ensure the maximum similarity of the central sources' coronagraphic PSFs for both stars.

We also adopt a PA constraint to ensure the offsets are applied in one direction aligned with the disk minor axis.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HR-4796A	RA: 12 36 0.9568 (189.0039867d) Dec: -39 52 10.59 (-39.86961d) Equinox: J2000	Proper Motion RA: -0.00483410349433121 sec of time/yr Proper Motion Dec: -0.023879999980636057 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[A stars]				
(2)	HR-4735	RA: 12 26 51.6779 (186.7153246d) Dec: -32 49 48.88 (-32.83024d) Equinox: J2000	Proper Motion RA: -9.028730739523227E-4 sec of time/yr Proper Motion Dec: -0.028128999929322163 arcsec/yr Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[B stars]				

Observation

Proposal 3087, Observation 1: HR-4796A offset-0

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(HR-4796A offset-0 (Obs 1)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(HR-4796A offset-0 (Obs 1)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HR-4796A	RA: 12 36 0.9568 (189.0039867d) Dec: -39 52 10.59 (-39.86961d) Equinox: J2000	Proper Motion RA: -0.00483410349433121 sec of time/yr Proper Motion Dec: -0.023879999980636057 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	true	SUB640A210R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	6	1	1	64.421	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	10	20	1	20	921.294	

PSF References	PSF offset-0 (Obs 4) (PSF Reference; Filters [F210M/F410M]) Additional Justification: false
Special Requirements	Aperture PA Range 110 to 124 Degrees (V3 109.98125598 to 123.98125598) Offset -0.004 arcsec, -0.011 arcsec No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible

Observation

Proposal 3087, Observation 2: HR-4796A offset-0.1arcsec

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(HR-4796A offset-0.1arcsec (Obs 2)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(HR-4796A offset-0.1arcsec (Obs 2)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HR-4796A	RA: 12 36 0.9568 (189.0039867d) Dec: -39 52 10.59 (-39.86961d) Equinox: J2000	Proper Motion RA: -0.00483410349433121 sec of time/yr Proper Motion Dec: -0.023879999980636057 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	true	SUB640A210R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	6	1	1	64.421	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	5	38	1	38	955.15	

PSF References	PSF offset-0.1arcsec (Obs 5) (PSF Reference; Filters [F210M/F410M]) Additional Justification: false
Special Requirements	Aperture PA Range 110 to 124 Degrees (V3 109.98125598 to 123.98125598) Offset -0.004 arcsec, 0.089 arcsec No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible

Observation

Proposal 3087, Observation 3: HR-4796A offset-0.15arcsec

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(HR-4796A offset-0.15arcsec (Obs 3)) Warning (Form): Science observations should be linked to at least one other compatible science observation by an Aperture PA Offset of 1-14 degrees

(HR-4796A offset-0.15arcsec (Obs 3)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	HR-4796A	RA: 12 36 0.9568 (189.0039867d) Dec: -39 52 10.59 (-39.86961d) Equinox: J2000	Proper Motion RA: -0.00483410349433121 sec of time/yr Proper Motion Dec: -0.023879999980636057 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[A stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	true	SUB640A210R	NONE

Confirmation

#	Conf. Readout Pattern	Conf. Groups/Int	Conf. Integrations/Exp	Conf. Total Integrations	Conf. Total Exposure Time	Conf. Total Dithers
1	RAPID	6	1	1	64.421	1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	4	45	1	45	942.736	

PSF References	PSF offset-0.15arcsec (Obs 6) (PSF Reference; Filters [F210M/F410M]) Additional Justification: false
Special Requirements	Aperture PA Range 110 to 124 Degrees (V3 109.98125598 to 123.98125598) Offset -0.004 arcsec, 0.139 arcsec No Parallel Attachments Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible

Observation

Proposal 3087, Observation 4: PSF offset-0

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(PSF offset-0 (Obs 4)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HR-4735	RA: 12 26 51.6779 (186.7153246d) Dec: -32 49 48.88 (-32.83024d) Equinox: J2000	Proper Motion RA: -9.028730739523227E-4 sec of time/yr Proper Motion Dec: -0.028128999929322163 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[B stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	false	SUB640A210R	9-POINT-CIRCLE

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	10	3	9	27	1243.747	

PSF References

PSF Reference: true

Special Requirements	
	Offset -0.004 arcsec, -0.011 arcsec Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible

Observation

Proposal 3087, Observation 5: PSF offset-0.1arcsec

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(PSF offset-0.1arcsec (Obs 5)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HR-4735	RA: 12 26 51.6779 (186.7153246d) Dec: -32 49 48.88 (-32.83024d) Equinox: J2000	Proper Motion RA: -9.028730739523227E-4 sec of time/yr Proper Motion Dec: -0.028128999929322163 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[B stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	false	SUB640A210R	9-POINT-CIRCLE

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	5	5	9	45	1131.098	

PSF References

PSF Reference: true

Special Requirements	
	Offset -0.004 arcsec, 0.089 arcsec Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible

Observation

Proposal 3087, Observation 6: PSF offset-0.15arcsec

Tue Feb 17 21:00:10 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRCam Coronagraphic Imaging

Diagnostics

(PSF offset-0.15arcsec (Obs 6)) Warning (Form): Specified offset will be added to offset from Coronagraphic Offsets file in the SOC PRD

(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	HR-4735	RA: 12 26 51.6779 (186.7153246d) Dec: -32 49 48.88 (-32.83024d) Equinox: J2000	Proper Motion RA: -9.028730739523227E-4 sec of time/yr Proper Motion Dec: -0.028128999929322163 arcsec/yr Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[B stars]

Acquisition

#	Target	Filter	Target Brightness	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	SAME	F210M	BRIGHT (ND Square)	MEDIUM8	33	1	1	59.883	275238

Template

Module	Occulting Mask	Obtain Astrometric Confirmation Images?	Subarray	Dither Pattern
A	MASK210R	false	SUB640A210R	9-POINT-CIRCLE

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	F210M	F410M	RAPID	4	5	9	45	942.736	

PSF References

PSF Reference: true

Special Requirements	
	Offset -0.004 arcsec, 0.139 arcsec Group Observations 1, 2, 3, 4, 5, 6, Non-interruptible